

A candidate counterexample to Informal Question (Section 2)

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

No: every such bijection satisfies $n \leq 1 + (k-1) + \dots + (k-1)^{(D-1)}$, where D is its inverse Lipschitz constant, and linear 2-Lipschitz tree maps attain the resulting bound. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Informal Question \(Section 2\)](#) from *Lipschitz bijections between boolean functions*.

1.1 Problem statement

Question. Does there exist a map ϕ from Dictator to XOR which is C -Lipschitz, where each output bit depends on at most k input bits, and such that ϕ^{-1} is $o(\log n / \log k)$ -Lipschitz as $n, k \rightarrow \infty$?

1.2 Catalog context

Theorem 1 shows that for general maps the inverse must be at least $\Omega(1/k) \cdot \log n$ -Lipschitz, while for linear maps the bound is $\Omega(1/\log k) \cdot \log n$ -Lipschitz. Rao and Shinkar’s construction achieves $\delta(2, k) = O(1/\log k)$ for linear maps, matching the linear-map lower bound, but a gap remains for general maps between $\Omega(1/k)$ and $O(1/\log k)$.

Notes. Stated in running prose as ‘This raises the following question:’ without a labelled theorem environment.

1.3 Claimed result

Model verdict: disproved (confidence: high).

No: every such bijection satisfies $n \leq 1 + (k-1) + \dots + (k-1)^{(D-1)}$, where D is its inverse Lipschitz constant, and linear 2-Lipschitz tree maps attain the resulting bound.

The model’s own caveat: Uses full-cube bijections and global k -junta locality; literature priority has not been independently checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A sharp negative answer

The gap closes on the lower-bound side. In fact, the forward Lipschitz assumption is unnecessary for the lower bound, and a rooted-tree construction gives the exact optimum.

The proof below does not rely on the previous attempt's localized-dependency estimate. It retains the elementary inverse-edge setup, verified below, but replaces dependency-graph packing by a bounded-branching count of possible target outputs.

2.1.1 1. Statement

Let $Q_n = \{0, 1\}^n$, equipped with Hamming distance. A Dictator-to-XOR bijection is a bijection

$$\phi : Q_n \longrightarrow Q_n$$

such that

$$\bigoplus_{a=1}^n \phi_a(x) = x_1 \quad \text{for every } x \in Q_n. \quad (1)$$

Assume that, for each output coordinate a , there is a set $P_a \subseteq [n]$, with $|P_a| \leq k$, such that ϕ_a depends only on the input coordinates in P_a . Write

$$D = \text{Lip}(\phi^{-1}).$$

On a Hamming cube, the Lipschitz constant equals the maximum stretch of an edge, so D is an integer.

For $k \geq 2$ and $d \geq 1$, put

$$N_k(d) = \sum_{r=0}^{d-1} (k-1)^r.$$

2.1.1.1 Theorem Every bijection satisfying (1) and the locality condition obeys

$$\boxed{n \leq N_k(D)}. \quad (2)$$

Conversely, whenever $n \leq N_k(d)$, there is a **linear, 2-Lipschitz** Dictator-to-XOR bijection whose output coordinates depend on at most k inputs and whose inverse is d -Lipschitz.

Consequently, the minimum possible inverse Lipschitz constant is exactly

$$\boxed{D_{\min}(n, k) = \begin{cases} n, & k = 2, \\ \left\lceil \frac{\log((k-2)n+1)}{\log(k-1)} \right\rceil, & k \geq 3. \end{cases}} \quad (3)$$

This remains the exact optimum if any fixed forward Lipschitz bound $C \geq 2$ is imposed.

In particular,

$$\boxed{D \geq 1 + \frac{\log n}{\log k}}. \quad (4)$$

Thus the inverse cannot be $o(\log n / \log k)$ -Lipschitz.

2.1.2 2. The counting argument

We prove a slightly stronger pointwise statement.

Fix an input $x \in Q_n$, and write $y = \phi(x)$. For each output coordinate a , define

$$x^{(a)} = \phi^{-1}(y \oplus e_a), \quad R_a = \{i : x_i \neq x_i^{(a)}\}.$$

The sets R_a are nonempty and pairwise distinct.

Equation (1) gives

$$x_1^{(a)} = \bigoplus_b (y \oplus e_a)_b = \left(\bigoplus_b y_b \right) \oplus 1 = x_1 \oplus 1.$$

Hence

$$1 \in R_a \quad \text{for every } a. \quad (5)$$

For a set $T \subseteq [n]$, let

$$x^T = x \oplus \bigoplus_{i \in T} e_i.$$

Thus $x^{R_a} = x^{(a)}$.

Fix an integer $d \geq 1$. For nonempty $T \subseteq [n]$, with $|T| \leq d$, define the family of target labels

$$A_d(T) = \{a : T \subseteq R_a, |R_a| \leq d\}.$$

2.1.2.1 Claim For every such T ,

$$|A_d(T)| \leq N_k(d - |T| + 1). \quad (6)$$

2.1.2.2 Proof We use induction on $d - |T|$.

If $|T| = d$, every $a \in A_d(T)$ must satisfy $R_a = T$. Since the R_a are distinct,

$$|A_d(T)| \leq 1 = N_k(1).$$

Now suppose $|T| < d$. Because $T \neq \emptyset$ and ϕ is injective,

$$\phi(x^T) \neq \phi(x).$$

Choose an output coordinate b on which these two images differ:

$$\phi_b(x^T) \neq \phi_b(x). \quad (7)$$

Since ϕ_b depends only on P_b , equation (7) implies

$$P_b \cap T \neq \emptyset, \quad \text{and therefore} \quad |P_b \setminus T| \leq k - 1. \quad (8)$$

Consider any target label $a \in A_d(T)$ with $a \neq b$. By the definition of R_a ,

$$\phi(x^{R_a}) = \phi(x) \oplus e_a.$$

Since $a \neq b$, it follows that

$$\phi_b(x^{R_a}) = \phi_b(x) \neq \phi_b(x^T). \quad (9)$$

The inputs x^{R_a} and x^T differ precisely on $R_a \setminus T$, because $T \subseteq R_a$. Locality and (9) therefore imply

$$P_b \cap (R_a \setminus T) \neq \emptyset.$$

Thus a belongs to $A_d(T \cup \{i\})$ for at least one $i \in P_b \setminus T$.

We have proved the covering relation

$$A_d(T) \subseteq \{b\} \cup \bigcup_{i \in P_b \setminus T} A_d(T \cup \{i\}). \quad (10)$$

The exceptional set $\{b\}$ has size one; it does not matter whether b actually belongs to $A_d(T)$.

Applying the induction hypothesis to every child set $T \cup \{i\}$, and using (8), gives

$$\begin{aligned} |A_d(T)| &\leq 1 + \sum_{i \in P_b \setminus T} |A_d(T \cup \{i\})| \\ &\leq 1 + (k-1)N_k(d - |T|) \\ &= N_k(d - |T| + 1). \end{aligned}$$

This completes the induction. $\square$

2.1.2.3 Applying the claim By (5), every inverse edge at y changes input coordinate 1. Therefore

$$A_d(\{1\}) = \{a : |R_a| \leq d\}.$$

Taking $T = \{1\}$ in (6), we obtain the pointwise bound

$$\boxed{\# \{a : d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq d\} \leq N_k(d).} \quad (11)$$

For $d = D$, all n inverse edges have stretch at most D . Equation (11) yields

$$n \leq N_k(D),$$

proving (2).

The essential distinction from the earlier packing argument is this: at a partial input change T , choose **one** currently changed output b . Its own target label accounts for at most one possibility. Every other target must add an input from the set $P_b \setminus T$, which has size at most $k-1$. This gives a branching factor $k-1$, without any dependence on the radius d .

2.1.3 3. Sharpness: the rooted-tree construction

Let T be a rooted tree with n vertices, root 1, maximum number of children at most $k-1$, and height at most $d-1$.

Such a tree exists whenever

$$n \leq 1 + (k-1) + \dots + (k-1)^{d-1} :$$

take the first n vertices in breadth-first order in the complete rooted $(k-1)$ -ary tree of height $d-1$.

Index both input and output coordinates by the vertices of T . Define

$$\phi_v(x) = x_v \oplus \bigoplus_{w: w \text{ is a child of } v} x_w. \quad (12)$$

We verify all the required properties.

2.1.3.1 Locality Each ϕ_v depends on its own input and on its children's inputs, hence on at most

$$1 + (k - 1) = k$$

input coordinates.

2.1.3.2 Dictator-to-XOR condition In the XOR of all outputs, the root input appears once. Every nonroot input appears twice: in its own output and in its parent's output. Consequently,

$$\bigoplus_v \phi_v(x) = x_1.$$

2.1.3.3 Bijectivity The inverse is

$$x_v = \bigoplus_{w \in \text{subtree}(v)} y_w, \quad y = \phi(x). \quad (13)$$

Indeed, summing (12) over the subtree rooted at v cancels every input in that subtree except x_v . Thus every input is recovered uniquely.

2.1.3.4 Forward Lipschitz constant Flipping the root input changes only the root output. Flipping a nonroot input x_v changes exactly the outputs indexed by v and its parent. Hence

$$\text{Lip}(\phi) \leq 2.$$

2.1.3.5 Inverse Lipschitz constant By (13), flipping output y_w changes exactly those inputs indexed by ancestors of w , including w . Its inverse stretch is therefore

$$\text{depth}(w) + 1 \leq d.$$

Thus

$$\text{Lip}(\phi^{-1}) \leq d.$$

This proves the converse assertion. Choosing the smallest d with $n \leq N_k(d)$ proves the exact optimum (3).

For completeness, when $n \geq 2$, a forward Lipschitz bound below 2 is impossible: flipping an input coordinate other than 1 preserves output parity, so injectivity forces the output Hamming distance to be a positive even integer. Therefore the construction also uses the smallest possible forward Lipschitz constant.

2.1.4 4. Consequences for the question

For $k = 2$, equation (2) is simply

$$n \leq D.$$

For $k \geq 3$,

$$n \leq \frac{(k-1)^D - 1}{k-2},$$

which gives the second line of (3).

A convenient uniform consequence follows from

$$\sum_{r=0}^{D-1} (k-1)^r \leq \sum_{r=0}^{D-1} \binom{D-1}{r} (k-1)^r = k^{D-1}.$$

Thus

$$n \leq k^{D-1}, \quad \text{so} \quad D \geq 1 + \frac{\log n}{\log k}.$$

This excludes

$$D = o\left(\frac{\log n}{\log k}\right)$$

along every sequence with $n, k \rightarrow \infty$.

The degenerate locality case $k = 1$ permits no such bijection for $n \geq 2$: in (10), $P_b \setminus T$ is empty, so taking $T = \{1\}$ gives $n \leq 1$.

Conclusion. Under the standard full-cube and global-junta definitions in the question, nonlinear maps offer no improvement in the optimal maximum inverse stretch: linear 2-Lipschitz rooted-tree maps are already exactly optimal. No additional conjecture or unproved mathematical step is used. The remaining caveat is literature priority, which has not been independently checked here.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	1812.09215__00
Catalog entry	Informal Question (Section 2)
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Lipschitz bijections between boolean functions
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/1812.09215__00/output.md
Independent review	None
arXiv	arXiv:1812.09215
Previous attempt	attacks/1812.09215__00: gpt-5.6-sol verdict partial
Model's caveats	Uses full-cube bijections and global k-junta locality; literature priority has not been independently checked.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/1812.09215__00/`.